

Programming Assignment 02

Control Flow and Basic Data Structures

Instructions

- Solve all questions using only the concepts discussed in class.
- Do not use external libraries.
- Write clean and well-commented code.
- Think about the algorithm before writing the code.
- You may solve each problem in multiple ways.

Programming Challenges

Problem 1. Write a program that accepts marks of five subjects from the user, stores them in a list, calculates the average, and prints the corresponding grade according to your own grading scheme.

Problem 2. Accept a sentence from the user and determine:

- Number of words
- Number of characters
- Number of vowels
- Number of consonants

Problem 3. Given a list of integers entered by the user, separate the numbers into two different lists containing even and odd numbers respectively.

Problem 4. Accept a list of integers and display

- Largest element
- Smallest element
- Sum of all elements
- Average

without using Python's built-in functions such as `max()`, `min()`, or `sum()`.

Problem 5. Write a program to determine whether a given string is a palindrome. Ignore spaces and case sensitivity.

Problem 6. Accept a tuple containing student marks. Display

- Highest mark
- Lowest mark
- Number of students scoring above the class average

Problem 7. Create a dictionary where the keys are student names and the values are their marks.

Allow the user to search for a student's marks by entering the student's name.

Problem 8. Write a program that counts the frequency of each character in a given string using a dictionary.

Example:

`banana`

should produce

`b : 1`

`a : 3`

`n : 2`

Problem 9. Accept two sets from the user and display

- Union
- Intersection
- Difference
- Symmetric Difference

Problem 10. Create a simple contact book using a dictionary.

The program should allow the user to

- Add a contact
- Search for a contact
- Update a contact
- Delete a contact
- Display all contacts

Use a menu-driven approach.

Problem 11. Generate the following pattern for an input value n .

Example ($n = 5$)

```
*
**
***
****
*****
```

Problem 12. Generate the following pattern.

Example ($n = 5$)

```
*****
****
***
**
*
```

Problem 13. Generate Floyd's Triangle for a given value of n .

Example

```
1
2 3
4 5 6
7 8 9 10
```

Problem 14. Given a list of integers, remove all duplicate values while preserving the original order of the elements.

Problem 15. Write a menu-driven program called **Student Record Management System**.

Each student record should contain

- Name
- Roll Number
- Course
- Marks

The program should allow the user to

- Add a student
- Search for a student
- Update student information
- Delete a student
- Display all students
- Exit

Store the records using appropriate Python data structures.

Think Before You Code!